Certifying Functional Dependencies over Latent Entity Identity: Proofs

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Code: `https://github.com/suanlab/SemFD`

Abstract

We give full proofs for the results of the paper *Certifying Functional Dependencies over Latent Entity Identity*. Value equality is defined by an oracle that maps each value to its entity, and a functional dependency is measured by its g_3 error on the instance that the oracle induces. We prove the soundness of the repair-witness certificate (with an explicit miss-mass term, which vanishes under inverse oracle access), its validity under data-dependent stopping and its per-sample oracle cost, a lower bound for certifiers without inverse access, certified closure under Armstrong's rules, eventual certification and the witness-slack bound, validity for predictable (self-refining) witnesses, bounds under updates and under a change of oracle, and the behavior of the pair-conditional and unconditional pair measures g_1^c and g_1 under inference. We also record the protocols of the two LLM-oracle experiments.

1 Setting

A relation r is a nonempty finite multiset of tuples over attributes $\mathcal{A}$. For each $A \in \mathcal{A}$ an *oracle* is a function $f_A : \mathrm{dom}(A) \to \mathcal{E}_A$; $a \equiv_A b$ iff $f_A(a) = f_A(b)$. The canonical instance $\hat{r}$ replaces each cell $t[A]$ by $f_A(t[A])$. For single attributes X, Y, $g_3(X \to Y) = \min\{|S|/|r| : r \setminus S \models X \to Y\}$, evaluated on $\hat{r}$ unless stated otherwise. Tuples are drawn uniformly at random (with replacement) from r.

Numbering. Theorems 1–3 and Proposition 1 have the same numbers as in the paper. Paper Proposition 2 (witness slack) is Proposition 4 here, paper Proposition 3 (eventual certification) is Proposition 3, paper Proposition 4 (certified closure) is Theorem 4, and paper Proposition 5 (oracle transfer) is Proposition 7. The self-refining witness is Proposition 5, updates are Proposition 6, the statements on g_1^c and g_1 are Propositions 8 and 9, and Proposition 10 bounds the samples of a simpler, checkpointed certifier (not the implemented one). Proposition 2 (unary supports) backs the shortest-path remark after paper Proposition 4.

Proposition 1 (Transitivity by construction)**.** *$\equiv_A$ is an equivalence relation. For a symmetric reflexive relation J on* $\mathrm{dom}(A)$*, the statements below hold with $\equiv_A$ replaced by the transitive closure J^*, and in general not for J itself.*

Proof. $\equiv_A$ is the kernel of a function, hence reflexive, symmetric and transitive. The proofs below use only these three properties of $\equiv$, which J^* has. Lemma 1 uses transitivity essentially, and the counterexample $a\,J\,b$, $b\,J\,c$, $\neg(a\,J\,c)$ breaks the chain argument. □

Definition 1 (Repair witness)**.** *$W = (\pi, rep, N, \prec)$ consists of*

- *a partition π of* $\mathrm{dom}(X)$ *into clusters;*

- *a map rep from clusters to* $\mathrm{dom}(Y)$*;*
- *for each* $a \in \mathrm{dom}(X)$ *a set* $N(a) \subseteq \mathrm{dom}(X)$*;*
- *a strict total order* $\prec$ *on clusters.*

Clusters c, c' conflict *if* $rep(c) \not\equiv_Y rep(c')$*. A tuple* t *with* $a = t[X]$*,* $c = \pi(a)$ *is* kept*, written* $t \in \mathcal{K}(W)$*, iff* (i) $t[Y] \equiv_Y rep(c)$ *and* (ii) *there is no* $b \in N(a)$ *with* $\pi(b) \succ c$*,* $b \equiv_X a$ *and* $\pi(b)$ *conflicting with* c*. Let* $Z(t)$ *hold iff some* $b \in \mathrm{dom}(X) \setminus N(a)$ *satisfies* $\pi(b) \succ c$*,* $b \equiv_X a$ *and* $\pi(b)$ *conflicts with* c*. Define* $\nu(N) = \Pr_t[Z(t)]$*.*

2 Soundness

Lemma 1. $K' = \{t \in \mathcal{K}(W) : \neg Z(t)\}$ *satisfies* $X \to Y$ *on* $\hat{r}$*.*

Proof. Let $t_1, t_2 \in \mathcal{K}'$ with $t_1[X] \equiv_X t_2[X]$, and let $c_i = \pi(t_i[X])$. We show $t_1[Y] \equiv_Y t_2[Y]$. *Case* $c_1 = c_2$*.* By (i), $t_1[Y] \equiv_Y rep(c_1) \equiv_Y t_2[Y]$, and transitivity gives the claim. *Case* $c_1 \neq c_2$ *and the clusters do not conflict.* Then $t_1[Y] \equiv_Y rep(c_1) \equiv_Y rep(c_2) \equiv_Y t_2[Y]$. *Case* $c_1 \neq c_2$ *and the clusters conflict.* W.l.o.g. $c_2 \prec c_1$. Put $b = t_1[X]$: then $\pi(b) = c_1 \succ c_2$, $b \equiv_X t_2[X]$, and c_1 conflicts with c_2. If $b \notin N(t_2[X])$ then $Z(t_2)$ holds, contradicting $t_2 \in \mathcal{K}'$. Otherwise (ii) fails for t_2, contradicting $t_2 \in \mathcal{K}(W)$. So this case cannot occur. □

Theorem 1 (Soundness)**.** *For every witness* W*:* $g_3(X \to Y) \leq \Pr_t[t \notin \mathcal{K}(W)] + \nu(N)$*.*

Proof. By Lemma 1, removing $S = r \setminus \mathcal{K}'$ yields an instance satisfying $X \to Y$. Hence $g_3 \leq |S|/|r| \leq \Pr[t \notin \mathcal{K}(W)] + \Pr[Z(t)]$. □

Corollary 1 (Inverse access)**.** *If* $N(a) \supseteq f_X^{-1}(f_X(a)) \cap \mathrm{dom}(X)$ *for every* a*, then* $\nu(N) = 0$ *and* $g_3 \leq \Pr[t \notin \mathcal{K}(W)]$*.*

Proof. Every $b \equiv_X a$ lies in $N(a)$, so $Z(t)$ never holds. □

Remark 1 (Composite left-hand sides)**.** *For* $X = \{A, B\}$*, let* $f_X(a, b) = (f_A(a), f_B(b))$*. This is a function, so all statements above apply verbatim with* $\mathrm{dom}(X)$ *the set of value pairs occurring in* r*. The inverse is* $f_X^{-1}(e_A, e_B) \cap \mathrm{dom}(X) = \{(a, b) \in \mathrm{dom}(X) : a \in f_A^{-1}(e_A),\, b \in f_B^{-1}(e_B)\}$*, so complete component inverses give a complete composite inverse and Corollary 1 applies. Calls are charged through the component caches.*

3 Estimation and cost

Let $B(t) = \mathbb{1}[t \notin \mathcal{K}(W)] \in \{0, 1\}$ and $\mu = \mathbb{E}[B]$. Draw $t_1, t_2, \ldots$ i.i.d. uniform. Let (L_n, U_n) be the predictable-plug-in empirical-Bernstein confidence sequence of Waudby-Smith and Ramdas [8] (Thm. 2 of arXiv:2010.09686v7), with running intersection, where each side is a one-sided sequence at level α; in particular $\Pr[\exists n : \mu \geq U_n] \leq \alpha$.

Theorem 2 (Certification)**.** *Fix* δ*, a declared* $\bar{\nu} \geq 0$ *and a cap* $n_{\max}$*. Stop at the first* $n \leq n_{\max}$ *with* $U_n < \delta - \bar{\nu}$ *(output* CERTIFIED*) or with* $L_n \geq \delta - \bar{\nu}$*, or at* $n_{\max}$ *(output* NOT CERTIFIED*). Then:*

1. *(Validity) If $\nu(N) \le \bar{\nu}$, then* $\Pr[\textsc{certified} \wedge g_3 > \delta] \le \alpha$. *Only the upper sequence carries the guarantee, so it is used one-sided at level α (with $\log(1/\alpha)$ in place of $\log(2/\alpha)$). The lower sequence serves only to stop early for futility, which cannot affect validity. A* NOT CERTIFIED *output carries no guarantee. Under inverse access the condition holds with $\bar{\nu} = 0$.*

2. *(Per-sample cost) Evaluating $B(t)$ costs at most $3 + 3\kappa$ oracle calls (fewer with caching), where $\kappa = \max_a |N'(a)|$ and $N'(a) = N(a)$, or $N(a) \cup (f_X^{-1}(f_X(a)) \cap \mathrm{dom}(X))$ under inverse access: two Y-calls, at most two X-calls per candidate and one initial X-call; inverse access adds one alias lookup.*

3. *(Total cost) The total is at most $n_{\max}(3+3\kappa)$ calls. This is independent of $|r|$; it is independent of $|\mathrm{dom}(X)|, |\mathrm{dom}(Y)|$ whenever κ is, i.e. for a fixed number of proxy neighbours and alias classes of bounded size.*

Proof. (1) On the event $\mu < U_n$ for all n, which has probability $\ge 1-\alpha$ by the one-sided guarantee, certification at n gives $g_3 \le \mu+\nu < U_n+\bar{\nu} < \delta$ by Theorem 1. Since the confidence sequence is time-uniform, the data-dependent stopping time does not affect this. (2) Condition (i) needs $f_Y(t[Y])$ and $f_Y(rep(c))$: 2 calls. Under inverse access, evaluating $f_X(a)$ and fetching its alias class cost at most two calls. Condition (ii) examines each $b \in N'(a)$. Clusters, priorities and representatives are known without the oracle, so only candidates with $\pi(b) \succ c$ and $rep(\pi(b)) \neq rep(c)$ as strings need calls: $f_X(a)$, $f_X(b)$ and $f_Y(rep(\pi(b)))$, at most 3 per neighbour, with $f_X(a)$ shared across them. More tightly, the inverse case costs at most $4 + 2\kappa \le 3 + 3\kappa$, since a complete in-data alias class contains a and hence $\kappa \ge 1$. Without inverse access the cost is at most $2 + 3\kappa$. Representatives that are equal as strings are oracle-equal. (3) Follows from (2). □

Remark 2 (Dependence on the gap)**.** *If $\mu = \delta - \bar{\nu} - \Delta$, certification requires $U_n < \delta - \bar{\nu}$. For i.i.d. Bernoulli observations, the variance-sensitive width heuristics in Section 3.3 of [8] motivate a sample scale $\tilde{O}\big(\mu \log(1/\alpha)/\Delta^2\big)$, with logarithmic factors suppressed. This is not an asymptotic equivalence or a proved stopping-time bound for our implementation: the default schedule also accumulates iterated-logarithmic terms. Proposition 3 proves eventual certification, and Proposition 10 gives a finite-checkpoint power bound for a different certifier.*

Remark 3 (Relation to Cormode et al. [1])**.** *With $\equiv$ observed, the audit (ii) is unnecessary, and* $\Pr[t \notin \mathcal{K}(W)]$ *is the standard modal-representative estimator of g_3 for a fixed partition. Our contribution is that the partition comes from an unreliable proxy. Without (ii) the estimator is unsound: two oracle-equal values in different clusters with conflicting representatives hide a violation from every per-cluster check.*

4 Limits without inverse access

A *forward-only* certifier may read r and any proxy information, and may call f_X and f_Y on values of its choice (adaptively, possibly randomized), but has no inverse access. It is *α-valid* if on every instance with $g_3(X \to Y) > \delta$ it outputs CERTIFIED with probability at most α.

Theorem 3 (Lower bound)**.** *Let $D \ge 4$, $1/D \le \delta < 1/4$ and $m = \lfloor \delta D \rfloor + 1$ (so $m \le 2\delta D$). Assume the oracle answers with* opaque labels*: fresh labels drawn uniformly at random without replacement from a large label space (so pairwise distinct), unknown to the certifier, i.e. identifiers that carry no information beyond equality; probabilities are over the certifier's randomness and the labels. There is an instance I_0 with $|r| = |\mathrm{dom}(X)| = D$ and $g_3(X \to Y) = 0$ such that every forward-only α-valid certifier making at most q oracle calls outputs* CERTIFIED *on I_0 with probability*

at most $\alpha + q^2 m/(D(D-1))$, where this and every probability below is over the certifier's internal randomness, the random labels and (in the proof) the random pair set M. Hence certifying I_0 with probability $\beta > \alpha$ requires

$$q \;\geq\; \sqrt{(\beta-\alpha)\,D(D-1)/m} \;\geq\; \sqrt{(\beta-\alpha)(D-1)/(2\delta)}.$$

This holds for any proxy information that is independent of which values are oracle-equal.

Proof. Let $r = \{(v_i, w_i)\}_{i \leq D}$ with distinct strings v_i and distinct Y-values w_i, and let f_Y be the identity. Oracle answers are generated lazily: the oracle keeps an infinite stream of fresh, pairwise distinct random labels.

- In I_0 every newly queried value receives the next fresh label. All values are distinct entities, so $g_3 = 0$.
- In I_M, for a set M of m disjoint pairs, a newly queried value receives its partner's label if its partner was queried before, and the next fresh label otherwise. Each pair is one entity with two distinct Y-values, so $g_3(I_M) = m/D > \delta$.

Draw M uniformly at random among all sets of m disjoint pairs, independently of everything else. Both instances get the same table and the same proxy information.

Run the certifier on both instances with the same internal randomness and the same label stream. Let E be the event that the run on I_M queries a value whose partner it queried earlier. Until E occurs, both runs receive the same fresh labels in the same order, so they ask the same queries and agree in every step. In particular, if E occurs, the query set Q_0 of the run on I_0 contains both endpoints of some pair of M.

The answers on I_0 do not depend on M, so Q_0 is independent of M. Each of the at most $\binom{q}{2}$ pairs inside Q_0 belongs to M with probability $m/\binom{D}{2}$, by symmetry, so $\Pr[E] \leq \binom{q}{2}\frac{2m}{D(D-1)} \leq \frac{q^2 m}{D(D-1)}$. Off E the two outputs coincide. Hence $\Pr_{I_0}[\textsc{cert}] \leq \mathbb{E}_M \Pr_{I_M}[\textsc{cert}] + \Pr[E] \leq \alpha + \Pr[E]$, where both sides average over the certifier's randomness and the label stream (and the left side does not depend on M), using α-validity on every I_M. □

Remark 4 (Scope)**.** *The opaque-label assumption matters. An oracle that answers with a canonical name that itself occurs in the data, as an LLM answering "Walmart" may, partially reveals the inverse, and the bound need not apply.*

Remark 5. *With inverse access the same instances are easy. $f_X^{-1}(f_X(v_i))$ reveals the partner of v_i, so for any partition that does not merge distinct entities (e.g. singletons) we have $B \equiv 0$ on I_0, and the upper bound U_n of Theorem 2 with the implemented schedule (proof of Proposition 3) drops below δ after at most $2(\log(1/\alpha) + 0.08)/\delta + 1$ samples, independently of D; the implementation, which checks after each batch of 100 samples, certifies within one batch of that point if it lies below $n_{\max}$. Indeed, with $B \equiv 0$ we have $\hat{\mu}_{t-1} = 1/(2t)$ for $t \geq 2$, so $t\hat{\sigma}_{t-1}^2 \leq \pi^2/24 < 0.42$ and $\lambda_t = 1/2$ whenever $\log(1+t) \leq 19\log(1/\alpha)$; the center is 0 and the half-width is at most $2(\log(1/\alpha) + (\log 2 - \frac{1}{2})\pi^2/24)/n < 2(\log(1/\alpha) + 0.08)/n$. This covers every practical δ (for $\alpha = 0.05$, all $n \leq \alpha^{-19} \approx 5 \cdot 10^{24}$). As $\delta \to 0$ at fixed α the bets eventually shrink and the count gains a $\sqrt{\log}$ factor; a constant-bet sequence keeps $O(\log(1/\alpha)/\delta)$. Inverse access, or an assumption about the proxy's recall that cannot be checked cheaply, is therefore what makes the cost independent of the domain size. Without either, the cost must grow at least like $\sqrt{D}$. The bound is modest at moderate scale: about 550 calls for $D = 29{,}894$ (GeoNames-16k city names), $\delta = 0.05$, $\beta - \alpha = 1/2$.*

In the hard instances the proxy cannot see the matching. This is the situation of world-knowledge aliases with no string similarity, such as Holland/Netherlands *or* Equate/Walmart*.*

The bound concerns oracle calls, not row samples. It complements the $\Omega(\sqrt{N})$ row-sampling bound of Kivinen and Mannila [3] and Cormode et al. [1], which assumes that equality is observed.

5 Certified closure

Lemma 2 (Subadditivity; Väänänen [7], rules A4/A6; see also [2])**.** *If φ is derivable from Σ' by Armstrong's rules, then on any instance $g_3(\varphi) \le \sum_{\sigma\in\Sigma'} g_3(\sigma)$.*

Proof. (Equivalently: monotonicity of the minimum-repair measure [4] plus a union of repairs.) Let S_σ be an optimal removal set for σ. Exact FDs are preserved under taking subinstances, so $r \setminus \bigcup S_\sigma$ satisfies every $\sigma \in \Sigma'$ and hence φ, by soundness of Armstrong's rules. □

Theorem 4 (Certified closure)**.** *Fix, before any sampling, a candidate set $\mathcal{C}$ and levels α_σ for $\sigma \in \mathcal{C}$, and assume $\nu_\sigma(N) \le \bar{\nu}_\sigma$ for every $\sigma \in \mathcal{C}$ (automatic under inverse access, with $\bar{\nu}_\sigma = 0$). Run the certificate on every $\sigma \in \mathcal{C}$ and let $\Sigma \subseteq \mathcal{C}$ be the certified ones, with final bounds U_σ. Then with probability $\ge 1 - \sum_{\sigma\in\mathcal{C}} \alpha_\sigma$, simultaneously for all φ and $\Sigma' \subseteq \Sigma$ with $\Sigma' \vdash \varphi$, we have $g_3(\varphi) \le \sum_{\sigma\in\Sigma'}(U_\sigma + \bar{\nu}_\sigma)$.*

Proof. By the union bound over $\mathcal{C}$, which is fixed in advance, all confidence sequences cover their means at all times with probability $\ge 1 - \sum_{\mathcal{C}} \alpha_\sigma$. On this event each $\sigma \in \Sigma$ has $g_3(\sigma) \le U_\sigma + \bar{\nu}_\sigma$ by Theorem 1, whatever data-dependent subset Σ turns out to be. Lemma 2 then bounds every derivation. Summing the α_σ only over the certified Σ would be invalid, because Σ is selected by the data. No independence between certificates is needed: they may share data and oracle caches. □

Proposition 2 (Unary supports)**.** *For unary Σ and unary $\varphi = X \to Y$ with $X \ne Y$, every support of φ contains a directed path from X to Y in the graph with edges Σ. Hence the minimum of $\sum_{\sigma\in P}(U_\sigma + \bar{\nu}_\sigma)$ over such paths (edge weights U_σ under inverse access, where $\bar{\nu}_\sigma = 0$), which a shortest-path algorithm computes, equals the minimum over all supports and is a valid bound. It need not be tight to $g_3(\varphi)$.*

Proof. A path is a transitivity derivation whose support is its edge set. Conversely, the attribute closure of $\{X\}$ under a set of unary FDs is the set of vertices reachable from X, so any support deriving Y contains such a path; since weights are nonnegative, dropping the other premises cannot increase the sum. The bound need not equal $g_3(\varphi)$. □

6 Predictable witnesses, updates and oracle changes

Proposition 3 (Eventual certification)**.** *Fix the witness and let $\mu = \mathbb{E}[B]$. If $\mu < \delta - \bar{\nu}$ (in particular if $g_3 + \epsilon_\pi + \epsilon_{rep} < \delta - \bar{\nu}$, Proposition 4), the certificate run without the cap $n_{\max}$ certifies with probability at least $1 - \alpha$.*

Proof. The B_t are i.i.d. in $\{0,1\}$ with mean μ (uniform sampling with replacement, fixed witness and oracles). The implementation uses $\lambda_t = \min\{\sqrt{2L/(\hat{\sigma}^2_{t-1} t \log(1+t))}, c\}$ with $L = \log(1/\alpha)$, $c = 1/2$ and the regularized variance $\hat{\sigma}^2_{t-1} = (1/4 + \sum_{i<t}(B_i - \hat{\mu}_i)^2)/t \in [1/(4t), 1]$. Hence $\lambda_t \ge \ell_t := \min\{\sqrt{2L/(t\log(1+t))}, c\}$, so $A_n = \sum_{t\le n}\lambda_t \to \infty$, and $\lambda_t \le \sqrt{8L/\log(1+t)} \to 0$; both hold surely, and A_t is $\mathcal{F}_{t-1}$-measurable.

Center. $M_n = \sum_{t\le n}\lambda_t(B_t - \mu)/A_t$ is a martingale with $\sum_t \mathbb{E}[(\lambda_t(B_t-\mu)/A_t)^2 \mid \mathcal{F}_{t-1}] \le \sum_t \lambda_t^2/A_t^2 \le 1 + c\sum_{t\ge 2}(A_t - A_{t-1})/(A_t A_{t-1}) \le 1 + c/\ell_1$, a deterministic constant, so M_n converges almost surely, and Kronecker's lemma gives $\sum_{t\le n}\lambda_t(B_t-\mu)/A_n \to 0$ almost surely.

Width. For $\lambda \le 1/2$, $4\psi_E(\lambda) = -\log(1-\lambda) - \lambda = \sum_{k\ge 2} \lambda^k/k \le \lambda^2/(2(1-\lambda)) \le \lambda^2$, and $(B_t - \hat{\mu}_{t-1})^2 \le 1$, so the half-width $(L + \sum_{t\le n} 4(B_t - \hat{\mu}_{t-1})^2 \psi_E(\lambda_t))/A_n \le (L + \sum_{t\le n} \lambda_t^2)/A_n \to 0$, since $A_n \to \infty$ and $\lambda_t \to 0$ (a weighted Cesàro mean of λ_t).

Hence $\limsup_n U_n \le \mu < \delta - \bar{\nu}$ almost surely, so $U_n < \delta - \bar{\nu}$ for some n. The run stops before that only for futility, when the lower sequence L_n reaches $\delta - \bar{\nu} > \mu$; the lower sequence is the mirror image of the upper one and a valid one-sided level-α sequence, so this happens with probability at most α. □

Proposition 4 (Witness slack)**.** *Let $\mu = \mathbb{E}[B]$ for a fixed witness. Call an X-entity e consistent if every cluster of π holding one of its values holds only values of e and the representatives of these clusters are all one Y-entity r_e; let n_e, $n_e(r_e)$ and m_e count e's tuples, those with Y-entity r_e, and those with e's most frequent Y-entity. Then $\mu \le g_3 + \epsilon_\pi + \epsilon_{rep}$ for any candidate sets, and $g_3 \le \mu$ under inverse access, with ϵ_π the fraction of tuples whose X-entity is not consistent and $\epsilon_{rep} = \sum_{e \text{ consistent}} (m_e - n_e(r_e))/N$. Both terms vanish if π groups* $\mathrm{dom}(X)$ *by entity and every representative is of the most frequent Y-entity.*

Proof. The lower bound is Theorem 1 with $\nu = 0$. Fix a consistent e and a tuple t with $t[X] = a$ of entity e, in cluster c. The candidates that condition (ii) checks are values oracle-equal to a, that is, values of e (all of them under inverse access, a subset otherwise); each lies in a cluster represented by an element of r_e, as is c, so no candidate has a conflicting representative and condition (ii) holds. Hence $B(t) = 1$ iff $t[Y]$ is not of entity r_e, for exactly $n_e - n_e(r_e)$ tuples of e. An optimal repair of $X \to Y$ on the canonical instance keeps, per X-entity, the tuples of one Y-entity and at best the m_e tuples of the most frequent one, so $g_3 N = \sum_e (n_e - m_e)$. Bounding $B(t) \le 1$ for the tuples of entities that are not consistent,

$$\mu N - g_3 N \le \sum_{e \text{ consistent}} (m_e - n_e(r_e)) + \sum_{e \text{ not consistent}} m_e \le \epsilon_{rep} N + \epsilon_\pi N.$$

If π groups values by entity and representatives are of the most frequent Y-entity, every entity is consistent with $n_e(r_e) = m_e$, so both terms are 0. □

Proposition 5 (Predictable witnesses)**.** *Let the witness W_n used for the n-th sample be a function of the earlier samples and of the oracle answers obtained before it, and let $\mu_n = \mathbb{E}[B_n \mid \mathcal{F}_{n-1}]$ be the conditional mean of $B_n = \mathbb{1}[t_n \notin \mathcal{K}(W_n)]$. Let $\tilde{U}_n$ be the upper bound of the confidence sequence at step n before the running intersection, so that $U_n = \min_{i\le n} \tilde{U}_i$. Under inverse access, $\mu_n \ge g_3$ for every n, and with probability at least $1-\alpha$, simultaneously for all n, $\tilde{U}_n > \sum_{i\le n} \lambda_i \mu_i / \sum_{i\le n} \lambda_i \ge g_3$, hence $U_n > g_3$. The running minimum U_n need not exceed the current weighted mean when the μ_i change (it is only an upper bound on g_3, which is fixed). Hence Theorem 2(1) holds for the self-refining witness of the system (every entity whose alias class an earlier batch fetched becomes its own cluster), including the variant that also uses alias classes fetched by earlier candidates through the shared cache: these answers precede every sample of the current candidate, which is drawn uniformly and independently of them.*

Proof. Given $\mathcal{F}_{n-1}$, W_n is fixed and t_n is uniform, so μ_n is the $\mathbb{E}[B]$ of a fixed witness and $\mu_n \ge g_3$ by Theorem 1 with $\nu = 0$. The predictable-plug-in empirical-Bernstein process of Waudby-Smith and Ramdas [8], $M_n = \prod_{i\le n} \exp\big(\lambda_i(\mu_i - B_i) - 4(B_i - \hat{\mu}_{i-1})^2 \psi_E(\lambda_i)\big)$, with predictable $\lambda_i \in [0, 1)$ and $\hat{\mu}_{i-1}$, satisfies $\mathbb{E}[M_n \mid \mathcal{F}_{n-1}] \le M_{n-1}$: the one-step inequality in their proof of Theorem 2 uses only that $B_i \in [0, 1]$ has conditional mean μ_i given $\mathcal{F}_{i-1}$ (their without-replacement empirical-Bernstein supermartingale, eq. (30) of arXiv:2010.09686v7, is the same construction with a time-varying conditional mean), not that the means are equal. Ville's inequality gives $\Pr[\exists n : M_n \ge 1/\alpha] \le \alpha$,

and on the complement, rearranging $\log M_n < \log(1/\alpha)$ gives $\sum_i \lambda_i \mu_i / \sum_i \lambda_i < \tilde{U}_n$ for all n. A weighted average of values $\geq g_3$ is $\geq g_3$, so $\tilde{U}_i > g_3$ for every i and their running minimum U_n exceeds g_3 as well. Only this bound on the fixed quantity g_3 survives the intersection, and it is all that Theorem 2(1) uses. □

Proposition 6 (Updates)**.** *Let r' arise from r ($|r| = N$) by deleting n_- and inserting n_+ tuples, with the oracle unchanged and $N - n_- + n_+ > 0$. Then $g_3^{\hat{r}'} \leq 1 - ((1 - g_3^{\hat{r}})N - n_-)/(N - n_- + n_+)$, and the right-hand side is nondecreasing in $g_3^{\hat{r}}$.*

Proof. An optimal repair of r keeps $(1-g_3^{\hat{r}})N$ tuples satisfying $X \to Y$ on $\hat{r}$. At least $(1-g_3^{\hat{r}})N - n_-$ of them are in r'; their canonical values are unchanged and FDs are closed under subsets, so they form a consistent subset of r' of that size. □

Substituting $U + \bar{\nu}$ for $g_3^{\hat{r}}$ yields a bound that holds on the event of the original certificate for every later version. The bound stays at most δ while $(\delta - U - \bar{\nu})N \geq \delta n_- + (1-\delta)n_+$. Re-certifying at version j uses a level α_j fixed in advance (we use $\alpha/2^{j+1}$ divided by the number of candidates), so all versions are covered with probability at least $1 - \sum_j \alpha_j$. Forward oracle answers stay valid across versions, but alias lists do not, because insertions add spellings: a re-run must fetch the alias class of every entity it touches again.

Proposition 7 (Oracle transfer)**.** *For oracles f, f', let m be the smallest fraction of tuples whose removal leaves, for each attribute of $X \cup Y$, a value set that f and f' partition identically. Then $|g_3^{\hat{r}_f} - g_3^{\hat{r}_{f'}}| \leq m$. For one attribute, $m = 1 - \max_M \sum_{(e,e') \in M} w(e, e')$ over matchings M between f-classes and f'-classes, where $w(e, e')$ is the fraction of tuples whose value lies in $e \cap e'$.*

Proof. The first claim: remove an optimal f'-repair and the m-set; the rest satisfies the FD under f' and is partitioned identically by f; then exchange the roles. The second: the kept values agree iff no two kept cells $e \cap e'$ share an f-class or an f'-class, i.e. the kept cells form a matching, and keeping a cell entirely is never worse. □

7 Why not the pair-conditional measure

Let $g_1^c(X \to Y) = \Pr[t_1[Y] \neq t_2[Y] \mid t_1[X] = t_2[X],\ t_1 \neq t_2]$ over ordered pairs of distinct tuples.

Proposition 8. *There are instances with $g_1^c(X \to Y) = 0$, $g_1^c(Y \to Z) \leq 2k/m$ and $g_1^c(X \to Z) = 1$ for any $m > k \geq 2$, k even.*

Proof. Use one Y-value y. Add $m - k$ tuples (x_i, y, z_0) with distinct x_i, and for $j = 1..k/2$ two tuples (p_j, y, z_j^a), (p_j, y, z_j^b) with fresh values.

- All tuples share Y, so no X-agreeing pair disagrees on Y: $g_1^c(X \to Y) = 0$.
- Among the $m(m-1)$ ordered Y-agreeing pairs, the Z-agreeing ones are at least the $(m-k)(m-k-1)$ pairs within the z_0 group, so $g_1^c(Y \to Z) \leq 1 - \frac{(m-k)(m-k-1)}{m(m-1)} \leq \frac{2k}{m}$.
- The only X-agreeing pairs are the p_j pairs, which disagree on Z: $g_1^c(X \to Z) = 1$.

□

Proposition 9. *The unconditional g_1 (the fraction of ordered pairs of distinct tuples that agree on the left-hand side and differ on the right-hand side) satisfies $g_1(X \to Z) \leq g_1(X \to Y) + g_1(Y \to Z)$, $g_1(XW \to YW) \leq g_1(X \to Y)$ and $g_1(X \to YZ) \leq g_1(X \to Y) + g_1(X \to Z)$.*

Proof. All three measures share the denominator, so it suffices to map violating pairs. A pair violating $X \to Z$ agrees on X and differs on Z; if it differs on Y it violates $X \to Y$, otherwise it agrees on Y and differs on Z, so it violates $Y \to Z$. A pair violating $XW \to YW$ agrees on X and W and differs on YW, hence on Y, so it violates $X \to Y$. A pair violating $X \to YZ$ differs on Y or on Z, so it violates $X \to Y$ or $X \to Z$. □

Proposition 10 (A checkpointed certifier with a sample bound)**.** *This concerns a simpler certifier than the one implemented (whose stopping rule is the empirical-Bernstein sequence of Theorem 2); we state it to show how the gap, the per-candidate level and the witness slack govern the sample count. Fix the witness, let $\mu = \mathbb{E}[B]$, $\tau = \delta - \bar{\nu}$, and let $\nu \le \bar{\nu}$. Choose checkpoints $n_1 < n_2 < \cdots$ and levels $a_j > 0$ with $\sum_j a_j \le \alpha$ (e.g. $n_j = n_1 2^{j-1}$, $a_j = \alpha/(j(j+1))$), and certify at the first j with $U_j := \bar{B}_{n_j} + \sqrt{\log(1/a_j)/(2n_j)} < \tau$, where $\bar{B}_{n_j}$ is the mean of the first n_j samples.*

1. *With probability at least $1-\alpha$, every certification it makes is correct: $g_3 \le \mu + \nu < \delta$.*
2. *If $\Delta = \tau - \mu > 0$ and $\sqrt{\log(1/a_j)/(2n_j)} + \sqrt{\log(1/\beta)/(2n_j)} < \Delta$, it certifies by checkpoint j with probability at least $1-\beta$, after at most n_j samples and $n_j(3+3\kappa)$ oracle calls.*
3. *For the schedule $n_j = n_1 2^{j-1}$, $a_j = \alpha/(j(j+1))$, fixed before sampling, let $n_j^* = (\sqrt{\log(1/a_j)} + \sqrt{\log(1/\beta)})^2/(2\Delta^2)$. Then the first j with $n_j > n_j^*$ has $n_j \le \max\{n_1, 2n_j^*\}$, and n_j^* grows only like $\log j$. (For other summable levels this can fail: if $a_1 = a_2$ are tiny and a_3 is large, the first such j can be 3 with $n_3 > \max\{n_1, 2n_3^*\}$.)*

By Proposition 4, $\Delta \ge \tau - g_3 - \epsilon_\pi - \epsilon_{rep}$, so a true dependency is certified cheaply when its distance to the tolerance exceeds the witness slack by a margin; under a family-wise budget α is replaced by $\alpha/|\mathcal{C}|$, which enters only logarithmically.

Proof. The B_i are i.i.d. in $\{0,1\}$ with mean μ. (1) By Hoeffding's inequality, $\Pr[\mu \ge U_j] \le a_j$ for each j, so with probability at least $1 - \sum_j a_j \ge 1 - \alpha$ every U_j exceeds μ; on this event a certification at j gives $\mu < U_j < \tau$, and Theorem 1 gives $g_3 \le \mu + \nu < \tau + \bar{\nu} = \delta$. (2) By Hoeffding's inequality, $\bar{B}_{n_j} < \mu + \sqrt{\log(1/\beta)/(2n_j)}$ with probability at least $1-\beta$, and then $U_j < \mu + \Delta = \tau$, so the certifier has stopped by checkpoint j (certified, or earlier). The condition is $n_j > n_j^*$. Each sample costs at most $3 + 3\kappa$ calls (Theorem 2). (3) In this schedule a_j decreases, so n_j^* increases; if the first j with $n_j > n_j^*$ exceeds 1, then $n_{j-1} \le n_{j-1}^* \le n_j^*$ and $n_j = 2n_{j-1} \le 2n_j^*$. Since $\log(1/a_j) = \log(1/\alpha) + \log(j(j+1))$, $n_j^* = O(\log(j/\alpha) + \log(1/\beta))/\Delta^2$. □

8 LLM oracle protocol (E8)

Model: Qwen2.5-7B-Instruct, Q4_K_M GGUF quantization (sha256 `65b8fcd9...a1423`), run with llama-cpp-python 0.3.36 on CPU (`n_ctx`=512, `temperature`=0, `max_tokens`=24, `seed`=0). For each normalized labeler value the model sees the most frequent raw spelling. The first line of the answer is lowercased and stripped of all non-alphanumeric characters, and the result is the entity label (so the oracle is a function). System prompt, verbatim:

> You normalize the names of companies that label drug products. Given one labeler name exactly as written in a product listing, reply with the canonical name of the company only: fix spelling and capitalization, drop legal suffixes (Inc, LLC, Ltd, Corp, Co), punctuation, store or division numbers, and 'dba'/'see also' remarks; if the listing names a store brand together with its owner, reply with the owner. Reply with the name only, no explanation.

Few-shot examples (synthetic firm names, given as prior chat turns):

- `user:` Acme Pharmaceuticals, Inc. → `assistant:` Acme Pharmaceuticals
- `user:` ACME PHARMACUETICALS LLC → `assistant:` Acme Pharmaceuticals
- `user:` Bestway Drug Co. (dba BestCare) → `assistant:` Bestway Drug
- `user:` Northwind Health 220, LLC → `assistant:` Northwind Health

All 6,653 labeler values were canonicalized once; the certification runs were replayed from these cached answers, each distinct queried value counting as one call. Code: `exp/e8_llm_oracle.py`.

9 LLM oracle protocol (E14, Tax city names)

Same model, runtime and settings as E8, except `max_tokens`=16. Data: a 10,000-row sample of the Raha Tax table (`tax(seed=0, n_rows=10000)`), whose city column has 6,500 normalized values. Each value is shown as its most frequent raw spelling, and the answer is turned into an entity label as in E8. System prompt, verbatim:

> You normalize United States city names taken from address records. Given one city name exactly as written, reply with the standard name of that city only: fix spelling, restore dropped letters, and write abbreviated words in full (ST = Saint, FT = Fort, MT = Mount, N = North, S = South, E = East, W = West, PT = Port, LK = Lake, SPGS = Springs, HTS = Heights, CTR = Center, VLG = Village, BCH = Beach). Reply in capital letters with the name only, no state and no explanation.

Few-shot examples (real city names, given as prior chat turns): `FT LAUDERDLE` → `FORT LAUDERDALE`; `Saint Paul` → `SAINT PAUL`; `CLRWATER BCH` → `CLEARWATER BEACH`; `N LITTLE ROCK` → `NORTH LITTLE ROCK`. All 6,500 values were canonicalized once and the certification runs replayed from the cache, as in E8. Code: `exp/e14_llm_tax_city.py`.

10 Live LLM timing and preprocessing notes (E19)

E19 uses the E8 model, prompt and seven natural NDC candidates, one process with 16 threads, three certification seeds and one resolve run. Its wall clock includes data loading, proxy construction and decisions, but excludes model loading. Answer caches start empty; the proxy uses cached embeddings. Normalized labels differ from the resolve arm on 13/3333, 10/3392 and 10/2385 queried names, respectively. Consequently the live decisions are comparisons to the resolve arm, not tests against a common fixed oracle. A pre-registered replay (E22, `exp/e22_fixed_oracle_replay.py`) fixes that oracle to the resolve arm's saved answers for all 6,653 names and reruns the certify arm for seeds 0–9 without calling the LLM: none of the ten runs made a false certification (its seven decisions share a table, oracle and cache, so the run is the unit: 0 of 10 bounds the per-run rate by 0.259, exact one-sided 95%), one of the 20 true-FD opportunities was missed (*labcode* → *labeler*, seed 3, $U = 0.0411$), and runs used 2,202–3,321 oracle calls. Computed offline from all saved labels (evaluation only), the exact decisions match the resolve arm's for all seven candidates and the miss mass of the neighbour-only witness is at most 0.0051, below the declared $\bar{\nu} = 0.01$; this holds for this mapping only. The E8 replay instead fixes one complete mapping. The absence of false certifications in E19 does not establish stationarity or validate its declared miss-mass bound. Scripts: `exp/run_e19.sh`, `exp/e19_end_to_end.py`.

The data loaders use string columns with `keep_default_na=False`; empty strings remain ordinary values unless a dataset-specific filter removes them. GeoNames drops ambiguous clean

city/admin names and reverts colliding injected aliases to their clean spelling. Tax aligns dirty and clean rows, strips leading zeros from ZIP codes, and uses city-name abbreviations and typographical variants. GPG drops empty company numbers and names associated with more than one company number, normalizes postcode spacing, and sorts SIC-code lists. The service documents Companies House as the source of private/voluntary-sector names and company numbers [5]; the register experiment therefore checks alias coverage, not independent annotation accuracy. These choices constrain generalization.

11 SEC EDGAR filer names (E25)

E25 was pre-registered after the manuscript's first freeze, in two steps: an entry probe with fixed pass criteria (`literature/prereg_rb_sec_probe_2026-11-03.md`) and, after it passed, the full run (`prereg_rb_sec_full_2026-11-03.md`). The table is the submissions file `sub.txt` of the SEC Financial Statement Data Sets [6], 2015q1–2024q4, one row per submission, with columns *name* (tolerant; oracle = the filer's CIK; inverse = the filer's names in the table), *cik*, *sic*, *countryba*, *stprba*, *cityba* and *fye* (exact). Names filed under two or more CIKs are dropped (1,617 rows of 61 names), leaving 273,832 rows, 13,110 filers and 16,401 names; name↔cik candidates hold by construction and are excluded, leaving 40 unary candidates, 8 of them true. 2,832 filers use several names (30% of the tuples); 3,596 of their 3,822 name pairs (94%) have Levenshtein ratio below 0.9, i.e. they are mostly renames.

The only candidate whose decision differs between strings and filers is *name* → *stprba*: $g_3 = 0.032$ on Layer-1 strings, 0.063 under the oracle, because the state recorded for one filer varies across its filings over the decade (moves or inconsistent entries). Over 3 seeds (false certifications / missed true FDs, summed):

method	false / missed	accepts *name* → *stprba*
TANE, raw strings	3 / 0	3/3
TANE, Layer-1 strings	3 / 0	3/3
proxy quotient ($\theta = 0.9$)	3 / 0	3/3
estimator without the audit	3 / 0	3/3
ours, complete inverse	0 / 3	0/3
planner	0 / 0	0/3

Ours rejected *name* → *stprba* with $U = 0.091$–0.104 and missed *name* → *sic* ($g_3 = 0.039$) in every seed, stopping at the 20,000-sample cap with $U = 0.061$–0.065 (witness slack, as for GPG). It used 21,510–21,782 calls against 16,401 for canonicalizing every name (1.31–1.33×); the planner used 22,762–22,831 (1.39×) and found all 8 true FDs. No derived bound was violated (0 of 114). With the E11 incomplete inverse (the certificate run as if complete, $\bar{\nu} = 0$): at $p = 0.5$, 0 false certifications, 1 miss, measured $\nu \leq 0.025$; at $p = 0$ (neighbour sets only), *name* → *stprba* was certified in all 3 seeds, with measured ν up to 0.048. A sensitivity check added after seeing that result (not pre-registered; `exp/e25_sec_nubar.py`) runs the neighbour-only mode with the paper's declared $\bar{\nu} = 0.01$: no false certification; *name* → *stprba* was rejected in 3/3 with $U = 0.045$–0.049, i.e. only because of the declared margin, while the measured ν exceeds it. Scripts: `exp/e25_sec_probe.py`, `exp/e25_sec_discovery.py`, `exp/e25_sec_nubar.py`; data: `data/sec/fetch.sh` (SEC requires a contact User-Agent) and the hashes in `results/icde27/DATA_SNAPSHOTS.txt`.

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